

LCC Pico Board Overview

Documentation, source and gerbers for boards may be found at: <https://github.com/bocabob/LCC-RPi-Pico-Board>

Evolution

The LCC Pico development boards evolved from LocoNet designs dating back to 2021. These early turntable boards used Arduino Nanos and depended on turnout commands for indexing to hard-coded stepper positions. Fascia panel input was desired so a keypad version was developed. This proved to be cumbersome, so a small touch screen was added with a Mega processor. The screen size needed to increase to nine inches for usability so the Raspberry Pi Pico processor was selected. For me, the Pico represents the 'Goldilocks' processor of very good function, technical stability, and the easy-to-use Arduino IDE. Moving to the Pico enabled LCC migration in 2024 when the Single Thread library began to support the RP2040 processor family. With the OpenLcbCLib, full protocol support was enabled.

Families

There have been two basic designs, the Node and the Stepper. The Stepper provides for a stepper motor driver and dedicates supporting GP pins. The Node provides 16 GP pins through two 10P IDC connectors. While it will work with either the Pico or Pico2, the latter is recommended due to enhanced capabilities.

I have put the stepper driver circuitry on a separate board which can be driven from the Node board. So there should not be a need to continue with a specific Stepper node board design.

Common Features

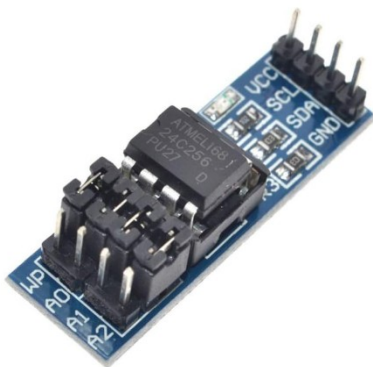
Power

The boards provide dual switching power output of 3.3V and 5V with input either from the LCC CAN bus or an external power source between 8 and 15 volts. The boards may be configured to inject power to the CAN bus or to isolate power for either or both of the CAN bus jacks. Auxiliary power outputs for both 3.3V and 5V are available on the board.

CAN Bus Control

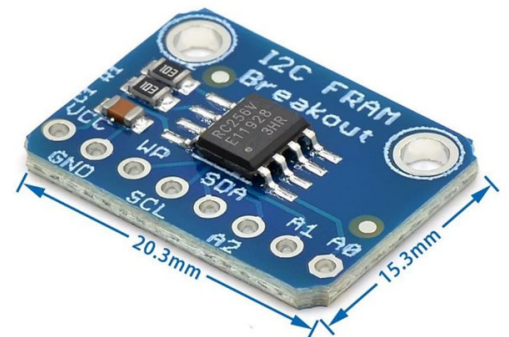
While there is an early design that uses the MCP251863 integrated CAN controller and transceiver, later designs settled on the MCP2518 and MCP2562 discrete hardware components (chips) for the LCC interface. Both work fine, it is just easier to hand solder the discrete chips.

Non-Volatile Memory (NVM)



I2C nonvolatile memory should be used to reduce strain on the Pico flash.

External NVM may be attached to either a I2C header, the FRAM Breakout header, or soldered down to the IC5 footprint. If either of the latter two choices are used, the I2C Address jumpers may be used to select non-default addresses or implement write protection. Otherwise, these would be functions of the module plugged into the I2C header.



A typical part would be a module utilizing an AT25C256 chip (left) or RC256V FRAM (right).

Parts compatible with the following should work for soldering to the footprint on the board.

- BR24G512F-3AGTE2 I2C BUS(2-Wre) 512K or other sizes with equivalent pinouts
- MB85RC256VPF-G-BCERE1 256kbit FRAM with I2C

I2C

I2C is used to address NVM. One or both of the available I2C channels on the Pico have headers on the board for external devices. Care should be taken should you attach peripherals to the same channel as the NVM lest there be interference. Some I2C devices will require 5 volts and so there is selection via jumper between the two available voltages.

Buttons

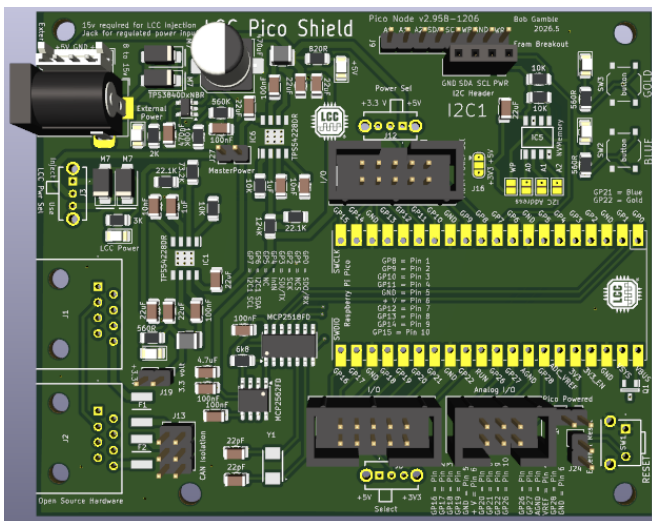
As there is no reset button on the Pico, one is provided on the board. There is also a header so that a button may be wired in for remote reset. There is also provision for blue and gold buttons should these be desired. Later designs have LEDs for these two buttons.

IDC Connections

GP pins and other connections are available via 10P IDC connectors. Pinouts may vary but are printed on the board and in the respective board user guide. Power is provided on pins 5 & 6 and is typically selectable except for the Turntable connector on the Stepper board.

The higher GPIO pins are used on the 2.8 Node board for the second IDC connector making the analog pins usable for node peripherals.

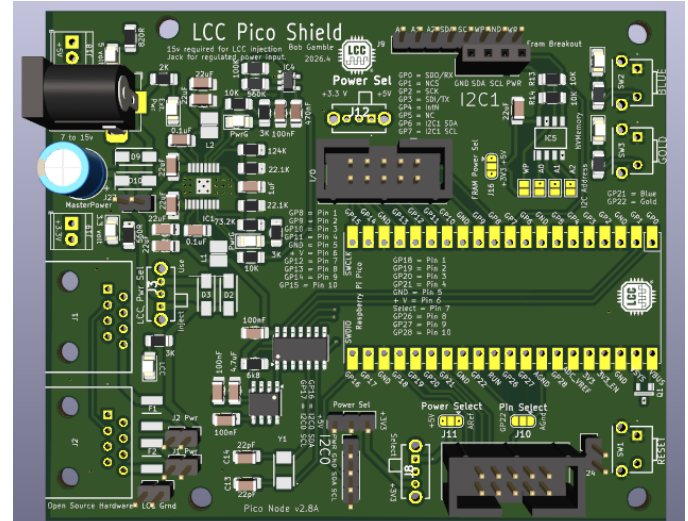
Node 2.95



The power circuit has been changed to two separate switching chips vs. the dual channel chip use on previous boards. The chips are identical in function. A board could be built just using the single 5-volt supply and using the Pico 3.3V output pin (through the provided jumper) to power the 3.3V for CAN and NVM circuits. Draw should not exceed 200-250mA in this configuration.

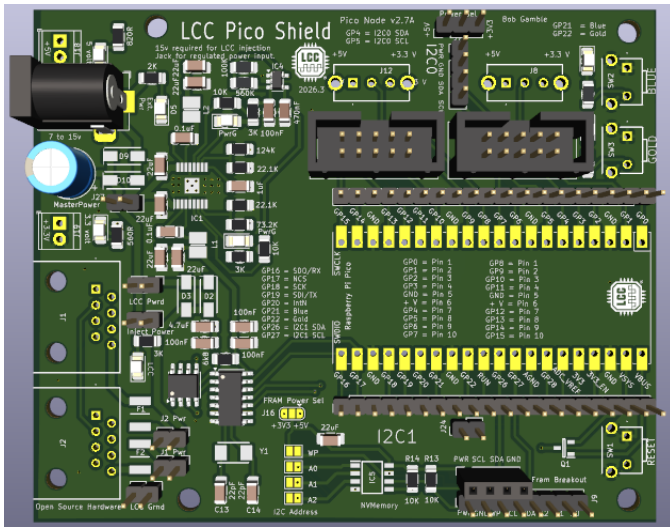
This node puts the analog GPIO pins on a separate connector to simplify the board. The I2C0 connector was removed as these pins are available in the second I/O connector to be used as desired.

Node 2.8



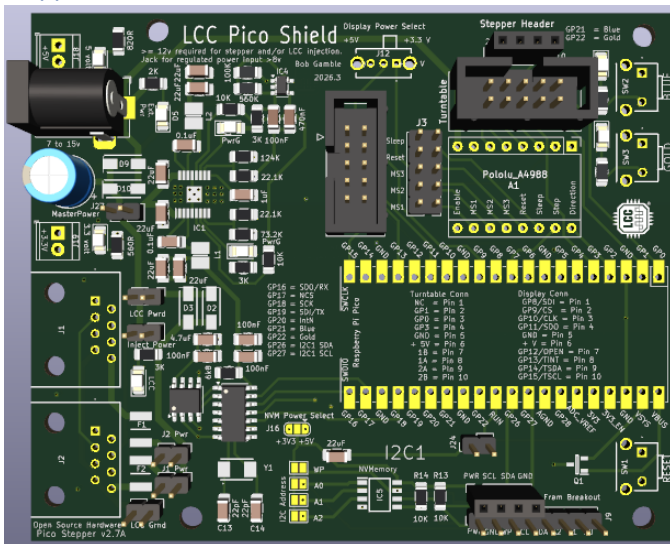
This node exposes the analog GPIO pins as an alternate implementation. Other than that, it has essentially the same functionality as Node 2.7.

Node 2.7



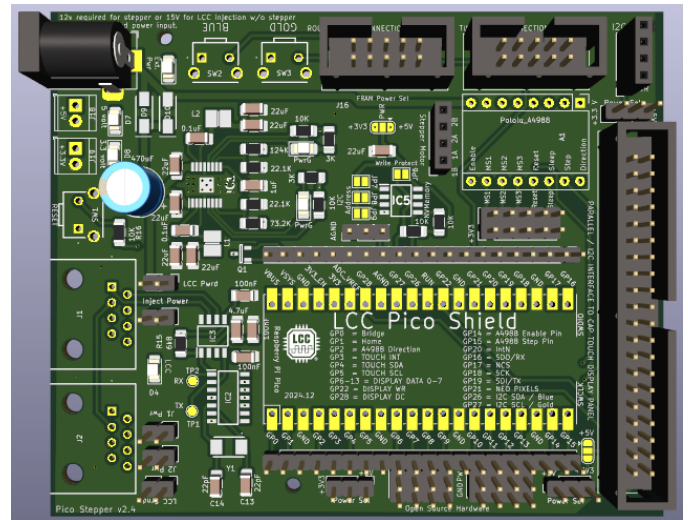
This node drives two 10P IDC connectors with a total of 16 GPIO pins. It is intended to drive servos and lights or to take inputs from detectors or buttons.

Stepper 2.7



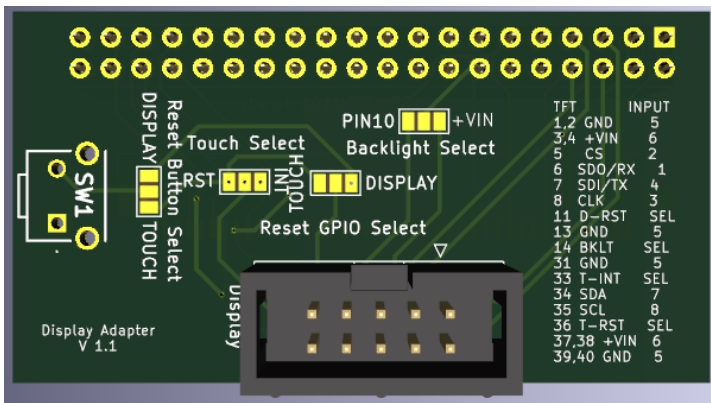
This node has provision for a stepper motor driver and is intended to control a turntable and SPI connected display.

Stepper 2.4

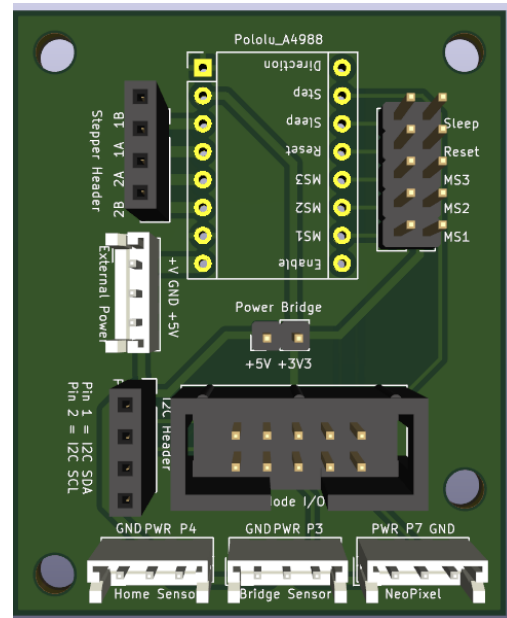


This node has provision for a stepper motor driver and is intended to control a turntable and parallel connected display.

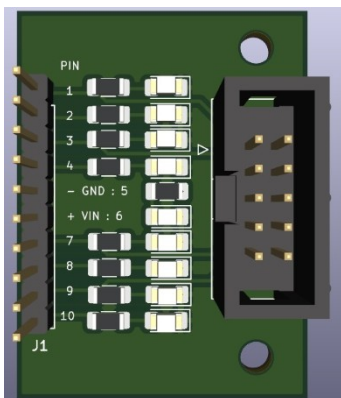
IDC Breakout Boards



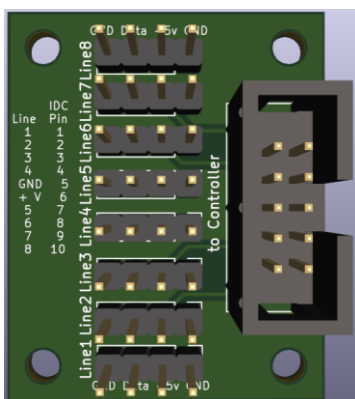
Display adapter attaches to a BuyDisplay ER-TFTM101 10.1" touchscreen for SDI display control and I2C touch. There is provision for alternate assignments of line 10 and a reset button.



This IDC breakout is intended to carry a stepper driver and provide connections to the stepper motor, sensors, NeoPixel bus, and an I2C connection. There is a separate power connection for the external power used for the stepper motor (12-15V) and for the +5V used for the hall sensors and NeoPixel bus. The IDC connector voltage should be set to 3.3V on the node side.



This IDC breakout has optional LEDs for testing.



This IDC breakout has power and ground pins around the GPIO pin to facilitate connections to inputs, servos, NeoPixels, or another device.

Board Pin Assignments

				2.5 Node board	2.6 Node board	2.7 Node board	2.8 Node board	2.95 Node board
1	gp0	SPI0 RX	I2C0 SDA	header pin	header pin	I/O-1:Pin1	ACAN_RX_PIN	ACAN_RX_PIN
2	gp1	SPI0 CSn	I2C0 SCL	header pin	header pin	I/O-1:Pin2	ACAN_CS_PIN	ACAN_CS_PIN
3	GND							
4	gp2	SPI0 SCK	I2C1 SDA	Npixel 1	Npixel 1	I/O-1:Pin3	ACAN_CSK_PIN	ACAN_CSK_PIN
5	gp3	SPI0 TX	I2C1 SCL	Npixel 2	Npixel 2	I/O-1:Pin4	ACAN_TX_PIN	ACAN_TX_PIN
6	gp4	SPI0 RX	I2C0 SDA	I2C0 SDA	I2C0 SDA	I/O-1:Pin7 / I2C0 SDA	ACAN_INT_PIN	ACAN_INT_PIN
7	gp5	SPI0 CSn	I2C0 SCL	I2C0 SCL	I2C0 SCL	I/O-1:Pin8 / I2C0 SCL	header pin	header pin
8	GND							
9	gp6	SPI0 SCK	I2C1 SDA	Npixel 3	Npixel 3	I/O-1:Pin9	I2C1 SDA	I2C1 SDA
10	gp7	SPI0 TX	I2C1 SCL	Npixel 4	Npixel 4	I/O-1:Pin10	I2C1 SCL	I2C1 SCL
11	gp8	SPI1 RX	I2C0 SDA	I/O-1:Pin10	I/O - Pin10	I/O-2:Pin1	I/O-1:Pin1	I/O-1:Pin1
12	gp9	SPI1 CSn	I2C0 SCL	I/O-1:Pin9	I/O - Pin9	I/O-2:Pin2	I/O-1:Pin2	I/O-1:Pin2
13	GND							
14	gp10	SPI1 SCK	I2C1 SDA	I/O-1:Pin8	I/O - Pin8	I/O-2:Pin3	I/O-1:Pin3	I/O-1:Pin3
15	gp11	SPI1 TX	I2C1 SCL	I/O-1:Pin7	I/O - Pin7	I/O-2:Pin4	I/O-1:Pin4	I/O-1:Pin4
16	gp12	SPI1 RX	I2C0 SDA	I/O-1:Pin4	I/O - Pin4	I/O-2:Pin7	I/O-1:Pin7	I/O-1:Pin7
17	gp13	SPI1 CSn	I2C0 SCL	I/O-1:Pin3	I/O - Pin3	I/O-2:Pin8	I/O-1:Pin8	I/O-1:Pin8
18	GND							
19	gp14	SPI1 SCK	I2C1 SDA	I/O-1:Pin2	I/O - Pin2	I/O-2:Pin9	I/O-1:Pin9	I/O-1:Pin9
20	gp15	SPI1 TX	I2C1 SCL	I/O-1:Pin1	I/O - Pin1	I/O-2:Pin10	I/O-1:Pin10	I/O-1:Pin10
21	gp16	SPI0 RX	I2C0 SDA	ACAN_RX_PIN	ACAN_RX_PIN	ACAN_RX_PIN	I2C0 SDA	I/O-2:Pin1
22	gp17	SPI0 CSn	I2C0 SCL	ACAN_CS_PIN	ACAN_CS_PIN	ACAN_CS_PIN	I2C0 SCL	I/O-2:Pin2
23	GND							
24	gp18	SPI0 SCK	I2C1 SDA	ACAN_CSK_PIN	ACAN_CSK_PIN	ACAN_CSK_PIN	I/O-2:Pin1	I/O-2:Pin3
25	gp19	SPI0 TX	I2C1 SCL	ACAN_TX_PIN	ACAN_TX_PIN	ACAN_TX_PIN	I/O-2:Pin2	I/O-2:Pin4
26	gp20		I2C0 SDA	ACAN_INT_PIN	ACAN_INT_PIN	ACAN_INT_PIN	I/O-2:Pin3	I/O-2:Pin7
27	gp21		I2C0 SCL	Blue Button	Blue Button	Blue Button	I/O-2:Pin4 & Blue Button	I/O-2:Pin8 & Blue Button
28	GND							
29	gp22			Gold Button	Gold Button	Gold Button	I/O-2:Pin7 & Gold Button	I/O-2:Pin9 & Gold Button
30	RUN			Reset Button / header	Reset Button / header	Reset Button / header	Reset Button / header	Reset Button / header
31	gp26	ADC0	I2C1 SDA	I2C1 SDA	I2C1 SDA	I2C1 SDA	I/O-2:Pin8	I/O-2:Pin10; I/O-3:Pin 1
32	gp27	ADC1	I2C1 SCL	I2C1 SCL	I2C1 SCL	I2C1 SCL	I/O-2:Pin9	I/O-3:Pin2
33	GND	A GND		header pin	header pin	header pin	I/O-2:Pin7 alternative	I/O-3:Pin3
34	gp28	ADC2		header pin	header pin	header pin	I/O-2:Pin10	I/O-3:Pin5
35		ADC_VREF		header pin	header pin	header pin	I/O-2:Pin6 alternative	I/O-3:Pin4
36	3v3_OUT							
37	3V3_EN							
38	GND							I/O-3:Pin6
39	VSYS							
40	VBUS							
	GND			I/O-1:Pin5	I/O-1:Pin5	I/O-1:Pin5	I/O-1:Pin5	I/O-1:Pin5
	Vselect1			I/O-1:Pin6	I/O-1:Pin6	I/O-1:Pin6	I/O-1:Pin6	I/O-1:Pin6
	GND					I/O-2:Pin5		
	Vselect2					I/O-2:Pin6		

				2.4 Stepper board	2.7 Stepper board	2.9 Stepper board
1	gp0	SPIO RX	I2C0 SDA	Bridge sensor	TT-Pin3 - Bridge sensor	ACAN_RX_PIN
2	gp1	SPIO CSn	I2C0 SCL	Home sensor	TT-Pin2 - Home sensor	ACAN_CS_PIN
3	GND					
4	gp2	SPIO SCK	I2C1 SDA	Stepper Direction	Stepper Direction	ACAN_CSK_PIN
5	gp3	SPIO TX	I2C1 SCL	Touch Interrupt	TT-Pin4 - NeoPixel data line	ACAN_TX_PIN
6	gp4	SPIO RX	I2C0 SDA	Touch SDA	header pin	ACAN_INT_PIN
7	gp5	SPIO CSn	I2C0 SCL	Touch SCL	header pin	TT-Pin1
8	GND					
9	gp6	SPIO SCK	I2C1 SDA	Display Data 0	Stepper Enable	I2C1 SDA
10	gp7	SPIO TX	I2C1 SCL	Display Data 1	Stepper Step	I2C1 SCL
11	gp8	SPI1 RX	I2C0 SDA	Display Data 2	I/O-2:Pin1 : SPI1 SDI	I/O-2:Pin1 : SPI1 SDI
12	gp9	SPI1 CSn	I2C0 SCL	Display Data 3	I/O-2:Pin2 : SPI1 CS	I/O-2:Pin2 : SPI1 CS
13	GND					
14	gp10	SPI1 SCK	I2C1 SDA	Display Data 4	I/O-2:Pin3 : SPI1 CLK	I/O-2:Pin3 : SPI1 CLK
15	gp11	SPI1 TX	I2C1 SCL	Display Data 5	I/O-2:Pin4 : SPI1 SDO	I/O-2:Pin4 : SPI1 SDO
16	gp12	SPI1 RX	I2C0 SDA	Display Data 6	I/O-2:Pin7 : Touch I2C0 SDA	I/O-2:Pin7 : Touch I2C0 SDA
17	gp13	SPI1 CSn	I2C0 SCL	Display Data 7	I/O-2:Pin8 : Touch I2C0 SCL	I/O-2:Pin8 : Touch I2C0 SCL
18	GND					
19	gp14	SPI1 SCK	I2C1 SDA	Stepper Enable	I/O-2:Pin9 : Touch INT	I/O-2:Pin9 : Touch INT
20	gp15	SPI1 TX	I2C1 SCL	Stepper Step	I/O-2:Pin10 : open	I/O-2:Pin10 : open
21	gp16	SPIO RX	I2C0 SDA	ACAN_RX_PIN	ACAN_RX_PIN	I2C0 SDA
22	gp17	SPIO CSn	I2C0 SCL	ACAN_CS_PIN	ACAN_CS_PIN	I2C0 SCL
23	GND					
24	gp18	SPIO SCK	I2C1 SDA	ACAN_CSK_PIN	ACAN_CSK_PIN	TT-Pin3 - Bridge sensor
25	gp19	SPIO TX	I2C1 SCL	ACAN_TX_PIN	ACAN_TX_PIN	TT-Pin2 - Home sensor
26	gp20		I2C0 SDA	ACAN_INT_PIN	ACAN_INT_PIN	TT-Pin4 - NeoPixel data line
27	gp21		I2C0 SCL	NeoPixel data line	Blue Button	Blue Button
28	GND					
29	gp22			Display WR	Gold Button	Gold Button
30	RUN			Reset Button	Reset Button / header	Reset Button / header
31	gp26	ADC0	I2C1 SDA	I2C SDA / Blue Button	I2C1 SDA	Stepper Enable
32	gp27	ADC1	I2C1 SCL	I2C SCL / Gold Button	I2C1 SCL	Stepper Step
33	GND	A GND			header pin	header pin
34	gp28	ADC2		Display D/C	header pin	Stepper Direction
35		ADC_VREF			header pin	header pin
36	3v3_OUT					
37	3V3_EN					
38	GND					
39	VSYS					
40	VBUS					
					TT-Pin1 - open	TT-Pin1 - open
					I/O-1:Pin5	I/O-1:Pin5
					I/O-1:Pin6	I/O-1:Pin6
					I/O-2:Pin5	I/O-2:Pin5
					I/O-2:Pin6	I/O-2:Pin6

Software

Example node applications may be found in:

https://github.com/bocabob/LCC_RPiPico_Turntable

https://github.com/bocabob/LCC_RPiPico_Roundhouse

https://github.com/bocabob/LCC_RPiPico_PixelLights

Board:

Must use the RP2040 board library by Earle Philhower: <https://arduino-pico.readthedocs.io/en/latest/install.html>

LCC support:

https://github.com/openlcb/OpenLCB_Single_Thread

<https://github.com/JimKueneman/OpenLcbCLib>

CAN Hardware:

ACAN2517 Arduino library.

External Storage:

I2C_EEPROM Arduino library by Rob Tillaart

FRAM_I2C Arduino library by Rob Tillaart

NeoPixel:

NeoPixelBus by Makuna Arduino library (preferred)

NeoPixelConnect Arduino library by Alan Yorinks.

Other preferred libraries for the RPi Pico:

Display

```
#include <TFT_eSPI_RA8876.h>    // display library [modified] https://github.com/bocabob/TFT\_eSPI\_RA8876
```

```
#include <RA8876_RP2040.h>    // display library [modified] https://github.com/bocabob/RA8876\_RP2040
```

```
#include "my_bb_captouch.h"    //capacitive touch driver [modified] https://github.com/bocabob/my\_bb\_captouch
```

Servos

```
#include <ServoEasing.hpp>    // great library for getting slow servo action, including bounces
```

```
#include <PCA9685_servo_driver.h>
```

```
#include <PCA9685_servo.h>
```

Stepper Motor

```
#include "AccelStepper.h"    // by Mike McCauley [modified]
```